# Erdős Problem 252

## Factorial tails, exact cancellation, and one surviving shift

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## The theorem concerns every divisor-power sum

For $n \geq 1$ and $k \in \mathbb{N} = \{0, 1, 2, \dots\}$, define

$$\sigma_k(n) = \sum_{d|n} d^k, \qquad \alpha_k = \sum_{n \geq 1} \frac{\sigma_k(n)}{n!}.$$

For example, the divisors of 6 are $1, 2, 3, 6$:

$$\sigma_0(6) = 4, \qquad \sigma_1(6) = 12, \qquad \sigma_2(6) = 50.$$

The series converges: $\sigma_k(n) \leq n^{k+1}$, and factorial growth wins.

**Main theorem**

$\alpha_k$ is irrational for every natural number $k$.

We explain the positive degrees first; $k = 0$ has a shorter ending.

# Earlier results reached the fourth power

| Exponent | Prior results |
| --- | --- |
| $k = 1, 2$ | Erdős–Kac; elementary tail arguments. |
| $k = 3$ | Schlage-Puchta (2006) and Friedlander–Luca–Stoiciu (2007), independently, using sieve methods. |
| $k = 4$ | Pratt (2022), using sieve methods and exponential sums. |

Earlier all-degree results assumed Schinzel's hypothesis H or a suitable prime-tuples conjecture.

**The argument here uses fixed coprime integers and exact cancellation.**

Its only prime-selection step is to choose one sufficiently large prime.

Source: Pratt (2022), introduction.

## Warm-up: factorials expose a rational denominator

Suppose $e = \sum_{m\geq 0} 1/m! = a/b$ with $a, b \geq 1$ integers. Then

$$b!\, e - b! \sum_{m\leq b} \frac{1}{m!} = b! \sum_{m>b} \frac{1}{m!} \ \in \mathbb{Z}.$$

Yet

$$0 < b! \sum_{m>b} \frac{1}{m!} < \sum_{j\geq 1} \frac{1}{(b+1)^j} = \frac{1}{b} \leq 1.$$

There is no integer strictly between 0 and 1.

**The principle we will reuse**

A sequence of integers tending to 0 is eventually exactly 0.

## Rationality would make all late tails integers

Fix $k$. Scale the tail after the $n$-th term:

$$T_n := n! \sum_{m>n} \frac{\sigma_k(m)}{m!}.$$

If $\alpha_k = a/b$ with $a \in \mathbb{Z}$ and $b \geq 1$, then for $n \geq b$,

$$T_n = \underbrace{\frac{n!a}{b}}_{\text{integer}} - \underbrace{\sum_{m=1}^{n} \sigma_k(m) \frac{n!}{m!}}_{\text{integer}}.$$

All later work begins with this consequence:

$$\alpha_k \in \mathbb{Q} \quad \Longrightarrow \quad T_n \in \mathbb{Z} \text{ for all sufficiently large } n.$$

## The original tails need not become small

The first terms are

$$T_n = \frac{\sigma_k(n+1)}{n+1} + \frac{\sigma_k(n+2)}{(n+1)(n+2)} + \cdots .$$

The divisor $n+1$ alone gives the exact lower bound

$$T_n \geq (n+1)^{k-1} \qquad (k \geq 1).$$

- For $k = 1$, the tail is at least 1; it cannot tend to 0.
- For $k \geq 2$, this lower bound tends to infinity.

**We need a small integer combination of tails, not a small individual tail.**

# The proof has four main moves

1. **Expand.** Separate a finite main term from an error smaller than $1/n$.
2. **Align.** Read several tails at compatible dilated indices.
3. **Cancel.** Finite differences remove all lower powers. The weighted integer becomes zero, forcing a remaining sum to tend to zero.
4. **Separate.** Two progression averages detect one surviving shift differently. They cannot both be zero.

**Factorials provide integrality; algebra cancels; averages give the contradiction.**

# 1. Reciprocal products have integer coefficients

Center each denominator at its largest factor $x$:

$$\frac{1}{x(x-1)} = \frac{1}{x^2} + \frac{1}{x^3} + \frac{1}{x^4} + \cdots,$$
$$\frac{1}{x(x-1)(x-2)} = \frac{1}{x^3} + \frac{3}{x^4} + \frac{7}{x^5} + \cdots.$$

In general, for $x > j$,

$$\frac{1}{x(x-1)\cdots(x-j)} = \sum_{i \geq j} \left\{ {i \atop j} \right\} \frac{1}{x^{i+1}}.$$

$\left\{ {i \atop j} \right\}$ counts partitions of $i$ objects into $j$ nonempty blocks. In particular, $\left\{ {i \atop j} \right\} = 0$ for $i < j$, and $\left\{ {j \atop j} \right\} = 1$.

**We keep only the powers through $x^{-(k+1)}$.**

# 1. The discarded part is smaller than $1/n$

With $j$ indexing the shifted argument and $i+1$ the denominator power,

$$T_n = \sum_{j=0}^{k} \sum_{i=j}^{k} \left\{ {i \atop j} \right\} \frac{\sigma_k(n+j+1)}{(n+j+1)^{i+1}} + E_n.$$

The error has two parts:

- the terms after the first $k+1$ terms of the actual tail;
- the higher reciprocal powers of each retained denominator.

Divisor pairing gives $\sigma_k(m) \le 64 m^k \sqrt{m}$. A geometric majorant and the finite Stirling remainder bound then give

$$0 \le E_n \le C_k n^{-3/2} \quad \text{for large } n, \qquad nE_n \longrightarrow 0.$$

Here $C_k$ is a fixed constant depending only on $k$.

## 2. Multiplicativity turns dilations into shifts

Choose a positive integer $p$ coprime to $1, \ldots, k+1$, and an offset $0 \le t < p$. For large $N \equiv t \pmod{p^2}$, put $n = (N-t)/p$.

$$p \mid n, \qquad \gcd(p, n+j+1) = \gcd(p, j+1) = 1 \quad (0 \le j \le k).$$

Define $r = (j+1)p - t$. Then $p(n+j+1) = N + r$, so

$$\sigma_k(p)\,\sigma_k(n+j+1) = \sigma_k(N+r).$$

Consequently,

$$\boxed{\sigma_k(p)\frac{\sigma_k(n+j+1)}{(n+j+1)^{i+1}} = p^{i+1}\frac{\sigma_k(N+r)}{(N+r)^{i+1}}}.$$

**The multiplier now appears as a polynomial factor $p^{i+1}$.**

## 2. Example: three tails share the same first shift

For $k = 1$, use three pairwise coprime multipliers, including the composite 9:

| | first tail | second tail | third tail |
|---|---|---|---|
| $p$ | 5 | 7 | 9 |
| $t$ | 0 | 2 | 4 |
| $n_p = (N - t)/p$ | $N/5$ | $(N-2)/7$ | $(N-4)/9$ |
| $p(n_p + 1)$ | $N + 5$ | $N + 5$ | $N + 5$ |
| $p(n_p + 2)$ | $N + 10$ | $N + 12$ | $N + 14$ |

Choose $N \equiv 0\ (25)$, $N \equiv 2\ (49)$, $N \equiv 4\ (81)$. The Chinese remainder theorem makes these conditions compatible.

**The first arguments coincide. The second arguments separate.**

## 3. Finite differences cancel polynomial factors

For every polynomial $P$ of degree at most $k$,

$$\sum_{a=0}^{k+1} b(a)P(a) = 0, \qquad b(a) = (-1)^{k+1-a}\binom{k+1}{a}.$$

For $k = 1$, this is the second-difference identity:

$$P(0) - 2P(1) + P(2) = 0.$$

In our example $p = 5 + 2a$, so

$$\underbrace{5 - 2 \cdot 7 + 9}_{0} \, \frac{\sigma_1(N+5)}{N+5} = 0.$$

**A shared arithmetic value can be arbitrary: its polynomial multiplier cancels exactly.**

## 3. Build one free coordinate for each lower shift

Let $e = (e_0, \dots, e_{k-1}) \in \{0, \dots, k+1\}^k$. Set

$$G = (k+2)^k - 1, \qquad D = G!, \qquad B = 1 + kDG,$$

and define

$$p_e = B + D \sum_{a<k} e_a (k+2)^a, \qquad t_e = D \sum_{a<k} (a+1) e_a (k+2)^a.$$

Their shift at position $j$ is

$$r_e(j) = (j+1)p_e - t_e = (j+1)B + D \sum_{a<k} (j-a) e_a (k+2)^a.$$

**For $j < k$, the shift ignores $e_j$: its coefficient is zero.**

But $p_e$ is affine in $e_j$, so $p_e^{i+1}$ is a polynomial in that coordinate.

## 3. The grid also guarantees the needed coprimality

Write $I_e = \sum_{a<k} e_a (k+2)^a$. Distinct vertices give distinct integers $I_e \in [0, G]$, and $p_e = B + DI_e \equiv 1 \pmod{D}$.

- **Pairwise coprime.** A common prime divisor $\ell$ of $p_e, p_f$ would divide $|I_e - I_f| \in [1, G]$, but not $D$. Then $\ell \le G$ forces $\ell \mid G! = D$, a contradiction.
- **Coprime to small shifts.** For $k \ge 1$, $k+1 \le G$; hence every $1, \dots, k+1$ divides $D$.
- **Positive arguments.** $t_e \le kDI_e \le kDG = B - 1 < p_e$, so every $r_e(j) > 0$.

**No multiplier needs to be prime, and no prime-pattern conjecture is used.**

## 3. A common progression preserves integrality

Use the product weights $w_e = \prod_{a<k} b(e_a) \in \mathbb{Z}$. Since the $p_e^2$ are pairwise coprime, choose

$$A \equiv t_e \pmod{p_e^2} \quad \text{for every } e, \qquad Q = \prod_e p_e^2.$$

Every $N = A + Qu$ satisfies the same congruences. For large $u$, let $n_e = (N - t_e)/p_e$ and form

$$\boxed{W(N) = \sum_e w_e \, \sigma_k(p_e) \, T_{n_e}}.$$

If $\alpha_k$ is rational, every $T_{n_e}$ is eventually integral:

$$W(A + Qu) \in \mathbb{Z} \quad \text{for all sufficiently large } u.$$

## 3. Only the top denominator power remains

After expansion and dilation, a typical contribution is

$$\left\{ {i \atop j} \right\} \sum_e w_e p_e^{i+1} \frac{\sigma_k(N + r_e(j))}{(N + r_e(j))^{i+1}}, \qquad 0 \le i, j \le k.$$

For every $i < k$:

- If $j < k$, hold all coordinates except $e_j$ fixed. The shifted value is constant in $e_j$; the polynomial degree is $i + 1 \le k$. Its weighted sum is zero.
- If $j = k$, the Stirling coefficient $\left\{ {i \atop k} \right\}$ is zero.

**All rows $i < k$ vanish identically. Only $i = k$ survives.**

This is an exact finite-sum identity, not an estimate.

## 3. Name what survives before taking any limits

Define the normalized divisor sum and the remaining coefficients:

$$\rho_k(m) = \frac{\sigma_k(m)}{m^k}, \qquad c_{e,j} = w_e p_e^{k+1} \begin{Bmatrix} k \\ j \end{Bmatrix}.$$

Then cancellation gives

$$W(N) = S(N) + E(N), \qquad S(N) := \sum_e \sum_{j=0}^{k} c_{e,j} \frac{\rho_k(N + r_e(j))}{N + r_e(j)}.$$

Here $E(N) = \sum_e w_e \sigma_k(p_e) E_{n_e}$ is a signed error. For each vertex, $N = p_e(n_e + 1) - r_e(0)$, so

$$NE_{n_e} = p_e(n_e + 1)E_{n_e} - r_e(0)E_{n_e} \longrightarrow 0.$$

The family is fixed and finite; therefore $NE(N) \to 0$.

## First limit: the weighted integer is eventually zero

All limits here are along the same progression $N = A + Qu$. From the divisor bound,

$$0 \leq \frac{\rho_k(m)}{m} \leq \frac{64}{\sqrt{m}} \longrightarrow 0.$$

There are only finitely many fixed coefficients and shifts, so

$$S(N) \longrightarrow 0, \qquad E(N) \longrightarrow 0.$$

Thus $W(N) = S(N) + E(N) \to 0$.

**Use integrality at exactly this point**

Eventually $W(N) \in \mathbb{Z}$ and $|W(N)| < 1$. Hence $W(N) = 0$, and consequently $S(N) = -E(N)$.

## Second limit: remove the remaining denominators

Define the survivor, with exactly the same coefficients and shifts:

$$R(N) := \sum_{e} \sum_{j=0}^{k} c_{e,j} \rho_k(N + r_e(j)).$$

For each fixed term, write $r = r_e(j)$ and $c = c_{e,j}$. Then

$$N \frac{c\rho_k(N+r)}{N+r} - c\rho_k(N+r) = -\frac{r\, c\rho_k(N+r)}{N+r} \longrightarrow 0.$$

Therefore $NS(N) - R(N) \to 0$. But the preceding slide gives $NS(N) = -NE(N) \to 0$.

**The consequence of rationality**

$\alpha_k \in \mathbb{Q}$ would force $R(A + Qu) \longrightarrow 0$.

# One shift has a coefficient that cannot cancel away

The shift

$$r_* = (k+1)B$$

occurs *only* at the zero vertex $e = 0$ and the last position $j = k$.

- For $j < k$, $r_e(j) \leq k(B + DG) = (k+1)B - 1$.
- For $j = k$, $r_e(k) = (k+1)B + D\sum_{a<k}(k-a)e_a(k+2)^a$. All added coefficients are positive, so equality requires $e = 0$.

Its coefficient is

$$c_* = w_0 B^{k+1} \left\{ {k \atop k} \right\} = B^{k+1} > 0, \qquad w_0 = (-1)^{k(k+1)} = 1.$$

Other shifts may repeat; this one does not.

## The survivor is concrete when $k = 1$

Recall $p_e = 5, 7, 9$, $t_e = 0, 2, 4$, and $w_e = 1, -2, 1$. The progression has modulus $Q = (5 \cdot 7 \cdot 9)^2 = 99225$.

$$W(N) = 6\,T_{N/5} - 16\,T_{(N-2)/7} + 13\,T_{(N-4)/9}.$$

The common first shift cancels because $5 - 14 + 9 = 0$. The second shifts have coefficients $25, -98, 81$, giving

$$\boxed{R(N) = 25\rho_1(N+10) - 98\rho_1(N+12) + 81\rho_1(N+14)}.$$

Rationality would force this expression to tend to 0 on the progression.

**The shift 10 stands alone, with positive coefficient 25.**

## 4. Divisor sums become divisor densities

Pair complementary divisors to rewrite

$$\rho_k(m) = \frac{\sigma_k(m)}{m^k} = \sum_{d \mid m} \frac{1}{d^k} \qquad (m > 0).$$

Along a progression $Qn + a$ with $Q, a > 0$, how often does a divisor occur?

| Example $Qn + a = 6n + 3$ | $d = 2$ | $d = 3$ | $d = 5$ | $d = 9$ |
|---|---|---|---|---|
| Density of $d \mid 6n + 3$ | 0 | 1 | $1/5$ | $1/3$ |

In general, with $g = \gcd(d, Q)$,

$$\text{density of } d \mid Qn + a = \begin{cases} g/d, & g \mid a, \\ 0, & g \nmid a. \end{cases}$$

The compatible congruence has one solution class modulo $d/g$.

## 4. The actual progression mean is explicit

For fixed $k \geq 1$ and $Q, a > 0$,

$$\frac{1}{X} \sum_{n=0}^{X-1} \rho_k(Qn + a) \longrightarrow \mu_Q(a) := \sum_{\substack{d \geq 1 \\ \gcd(d,Q) \mid a}} \frac{\gcd(d, Q)}{d^{k+1}} \geq 1.$$

Why may we sum the individual divisor densities? Counted values $Qn + a$ are distinct positive multiples of $d$, so for $X \geq 1$,

$$\frac{1}{d^k} \frac{\#\{n < X : d \mid Qn + a\}}{X} \leq \frac{QX + a}{X\, d^{k+1}} \leq \frac{Q + a}{d^{k+1}}.$$

This bound is summable in $d$. Dominated convergence applies.

**This includes $k = 1$: the majorant is a multiple of $d^{-2}$.**

## 4. A fresh prime changes the mean by one factor

Take a prime $L \nmid Q$ and refine the progression to $Q(Ln+v)+a$. Its starting point $a' = Qv + a$ is congruent to $a$ modulo $Q$.

$$\boxed{\mu_{QL}(a') = \mu_Q(a)\left(1 - L^{-(k+1)} + \mathbf{1}_{[L|a']} L^{-k}\right)}.$$

The indicator $\mathbf{1}_{[L|a']}$ is 1 if $L \mid a'$, and 0 otherwise.

- The terms with $L \nmid d$ contribute $\mu_Q(a)(1 - L^{-(k+1)})$.
- Write the other terms as $d = Ld'$. They contribute $\mu_Q(a)L^{-k}$ if $L \mid a'$, and zero otherwise.

**There is a common contribution, plus an extra term when the prime divides the offset.**

## 4. Choose one residue to miss and one to isolate

Write $i=(e,j)$, $r_i=r_e(j)$ and $c_i=c_{e,j}$, so $R(N)=\sum_i c_i\rho_k(N+r_i)$. Choose a prime $L>\max(Q,\max_i r_i)$. Since $\gcd(Q,L)=1$, choose:

| **Miss every shift** | **Hit only the isolated shift** |
|---|---|
| $Qv_0+A\equiv 0 \pmod{L}$ | $Qv_1+A+r_*\equiv 0 \pmod{L}$ |
| $Qv_0+A+r_i\equiv r_i \pmod{L}$ | $Qv_1+A+r_i\equiv r_i-r_* \pmod{L}$ |
| Never zero: $0<r_i<L$. | Zero exactly when $r_i=r_*$. |

These give two subprogressions of $N=A+Qu$. There is exactly one term at $r_*$, with coefficient $c_*>0$.

# 4. The two averages cannot both vanish

If $R(A+Qu) \to 0$, its mean on *each* positive-step subprogression is zero. But the prime-refinement formula gives the two means

$$m_0 = (1 - L^{-(k+1)}) \sum_i c_i \mu_Q(A + r_i),$$
$$m_1 = m_0 + c_* \mu_Q(A + r_*) L^{-k}.$$

Every offset $A + r_i$ is positive, so the mean theorem applies. Their difference satisfies

$$m_1 - m_0 = c_* \mu_Q(A + r_*) L^{-k} > 0,$$

because $c_* > 0$, $\mu_Q(A + r_*) \geq 1$, and $L^{-k} > 0$.

**Contradiction**

Both means would be zero. Hence $\alpha_k$ is irrational for every $k \geq 1$.

## Degree zero returns to the short tail argument

For $k = 0$, the first denominator term and the same error estimate give

$$T_n = \frac{\sigma_0(n+1)}{n+1} + E_n.$$

The divisor count is positive and at most $64\sqrt{n+1}$, so

$$0 < T_n \le \frac{64}{\sqrt{n+1}} + O(n^{-3/2}) \longrightarrow 0.$$

Rationality would make these tails integers for all sufficiently large $n$. That is impossible once $0 < T_n < 1$.

**The separate zero-degree argument completes every natural exponent.**

## The formal artifact proves the original series

$$\forall k \in \mathbb{N}, \qquad \mathsf{Irrational}\left(\sum_{n=0}^{\infty} \frac{\mathsf{ArithmeticFunction.sigma}(k, n)}{n!}\right).$$

The $n = 0$ term is zero. The statement audit also checks the expanded divisor sum, summability, and the equivalent series starting at $n = 1$.

- Lean 4.33.1, pinned Mathlib; full build and a fresh trust-zero build.
- Statement and axiom audits; fresh replay using Lean's kernel.
- Exactly `propext`, `Classical.choice`, and `Quot.sound`.

Proof module: **834 substantive lines; 1,120 total lines**. The one-line root import adds one to each count.

Source: Lean source and reproduction commands.

# Three ingredients resolve the contradiction

1. **Factorials turn rationality into integrality.** Every sufficiently late scaled tail would be an integer.
2. **A fixed dilation grid creates exact cancellation.** The weighted integer tends to zero, leaving one identifiable survivor.
3. **Progression means detect that survivor.** A fresh prime separates two averages that rationality would force to agree.

### Conclusion

For every natural $k$, $\sum_{n\geq 1} \sigma_k(n)/n!$ is irrational.

# Appendix: the finite Stirling bound used in Lean

Put

$$D_{j+1}(x) = \frac{1}{x(x-1)\cdots(x-j)}, \quad P_{j,L}(x) = \sum_{i=0}^{L-1} \left\{ {i \atop j} \right\} x^{-(i+1)}, \quad F_{j,L}(x) = D_{j+1}(x) - P_{j,L}(x).$$

The exact recurrence is

$$F_{j+1,L+1}(x) = \frac{F_{j,L}(x) + (j+1)F_{j+1,L}(x)}{x}.$$

For $j+1 \le H \le x$, induction on $L$ gives

$$0 \le F_{j,L}(x) \le H^L / x^{L+1}.$$

At $L = 0$, $0 < D_{j+1}(x) \le 1/x$. In the induction step, the weighted sum of two errors bounded by $u$ is at most $(j+2)u \le Hu$. The $j = 0$ error is zero after the first order.

**This proves the truncated expansion directly, with an explicit error.**

## Appendix: one numerator bound controls both errors

For $x \geq n+1$, the divisor bound gives

$$\sigma_k(x) \leq 64x^k\sqrt{x} \leq \frac{64x^{k+1}}{\sqrt{n+1}}.$$

For a retained term, multiply by the Stirling remainder:

$$0 \leq \sigma_k(x)F_{j,k+1}(x) \leq \frac{64(k+1)^{k+1}}{x\sqrt{n+1}} \leq \frac{64(k+1)^{k+1}}{(n+1)^{3/2}}.$$

For the omitted terms, factorial denominators yield a geometric majorant; their total is at most $64(k+1)^{k+1}/(n\sqrt{n+1})$ for $n > 0$. There are $k+1$ retained errors. Hence, for sufficiently large $n$,

$$E_n \leq \frac{64(k+1)^{k+1}}{n\sqrt{n+1}} + \frac{64(k+1)^{k+2}}{(n+1)^{3/2}}.$$

This is the explicit $O_k(n^{-3/2})$ estimate.

## Appendix: a compact notation guide

| | |
|---|---|
| $k \geq 1$ | fixed divisor-power exponent in the grid argument |
| $e \in \{0, \dots, k+1\}^k$ | one vertex of the finite grid |
| $p_e, \ t_e$ | multiplier and offset for that vertex |
| $r_e(j) = (j+1)p_e - t_e$ | shift of the $(j+1)$-st tail term |
| $w_e = \prod_{a<k} (-1)^{k+1-e_a} \binom{k+1}{e_a}$ | integer finite-difference weight |
| $A, \ Q = \prod_e p_e^2$ | CRT starting point and progression modulus |
| $n_e = (N - t_e)/p_e$ | tail index, for large $N = A + Qu$ |
| $c_{e,j} = w_e p_e^{k+1} \left\{ {k \atop j} \right\}$ | coefficient after cancellation |
| $r_* = (k+1)B, \ c_* = B^{k+1}$ | the isolated shift and its coefficient |
| $L, \ v_0, \ v_1$ | fresh prime and the two separating residues |

# Appendix: four quantities with different roles

$$T_n = n! \sum_{m>n} \frac{\sigma_k(m)}{m!}$$ one scaled tail,

$$W(N) = \sum_e w_e \sigma_k(p_e) T_{n_e}$$ the weighted integer,

$$S(N) = \sum_e \sum_{j=0}^{k} \frac{c_{e,j}\rho_k(N + r_e(j))}{N + r_e(j)}$$ the surviving divided sum,

$$R(N) = \sum_e \sum_{j=0}^{k} c_{e,j}\rho_k(N + r_e(j))$$ the raw survivor.

The normalized divisor sum is $\rho_k(m) = \sigma_k(m)/m^k$. The logical chain is

$$W = S + o(1/N), \quad W \in \mathbb{Z}, \quad W \to 0 \implies W = 0 \text{ eventually} \implies NS \to 0 \implies R \to 0.$$

The last implication uses $NS - R \to 0$, not the false identity $NS = R$.

# Sources and the complete proof

- **Formal proof and full mathematical exposition**
  Tokengrinder, *Erdős 252* (2026). github.com/tokengr1nder/Erdos252
  `PROOF.pdf` contains all 94 named proof results and their proofs, 40 definitions, and the six statement-audit results.

- **Historical context**
  K. Pratt, *The irrationality of a divisor function series of Erdős and Kac* (2022), introduction.
  arXiv:2209.11124.

- J. B. Friedlander, F. Luca, M. Stoiciu, *On the irrationality of a divisor function series*, Integers 7 (2007), A31.
  Original paper.

- **Problem catalogue**
  T. F. Bloom, Erdős Problem 252.